LAPORAN PRAKTIKUM

POSTTEST (7)

ALGORITMA PEMROGRAMAN LANJUT

Disusun oleh:

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PROGRAM STUDI INFORMATIKA

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SAMARINDA

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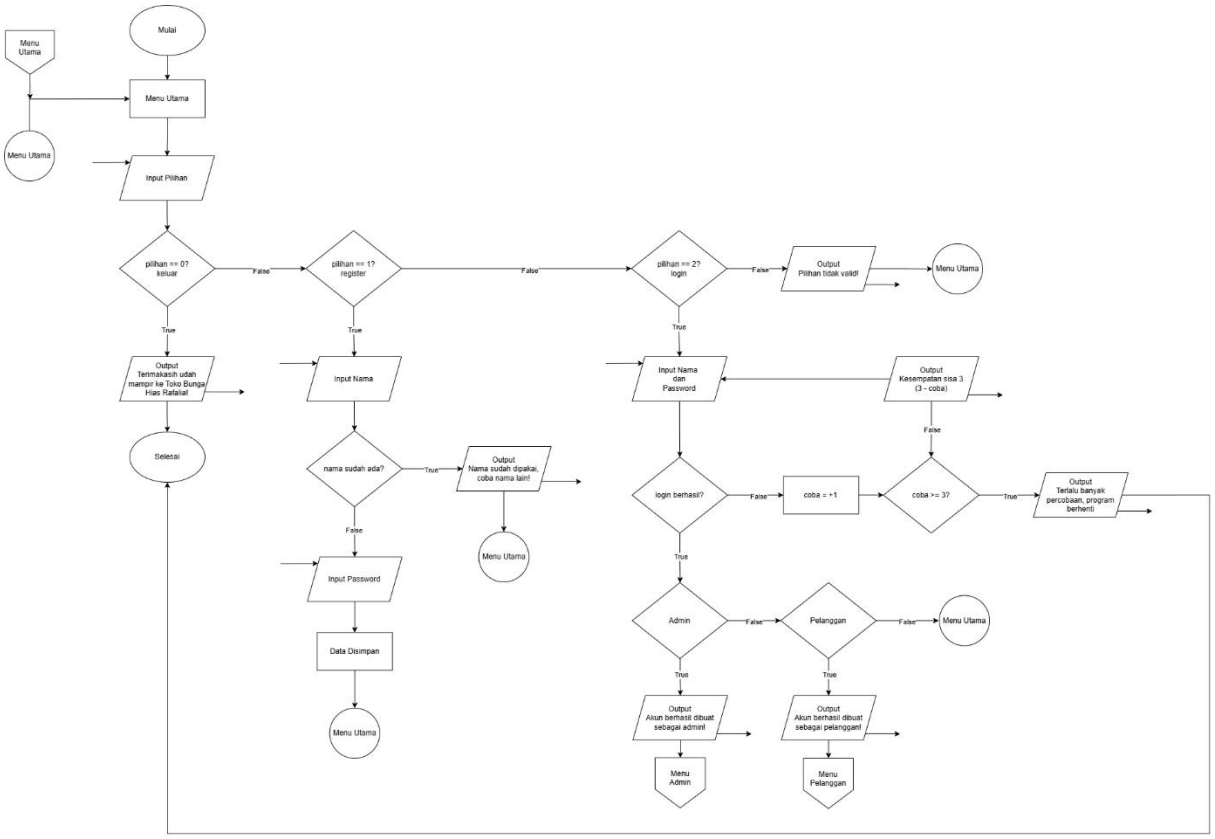
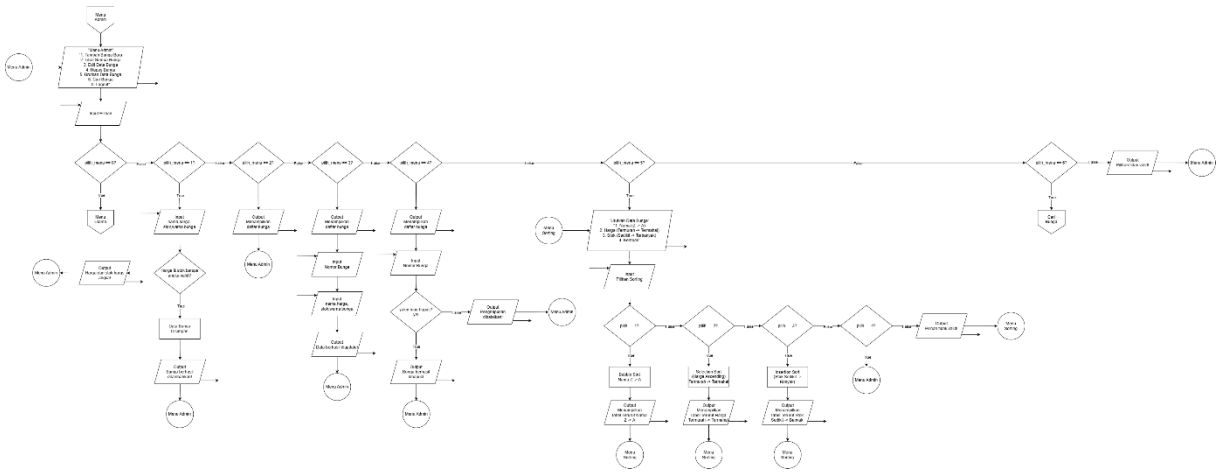
1. Flowchart

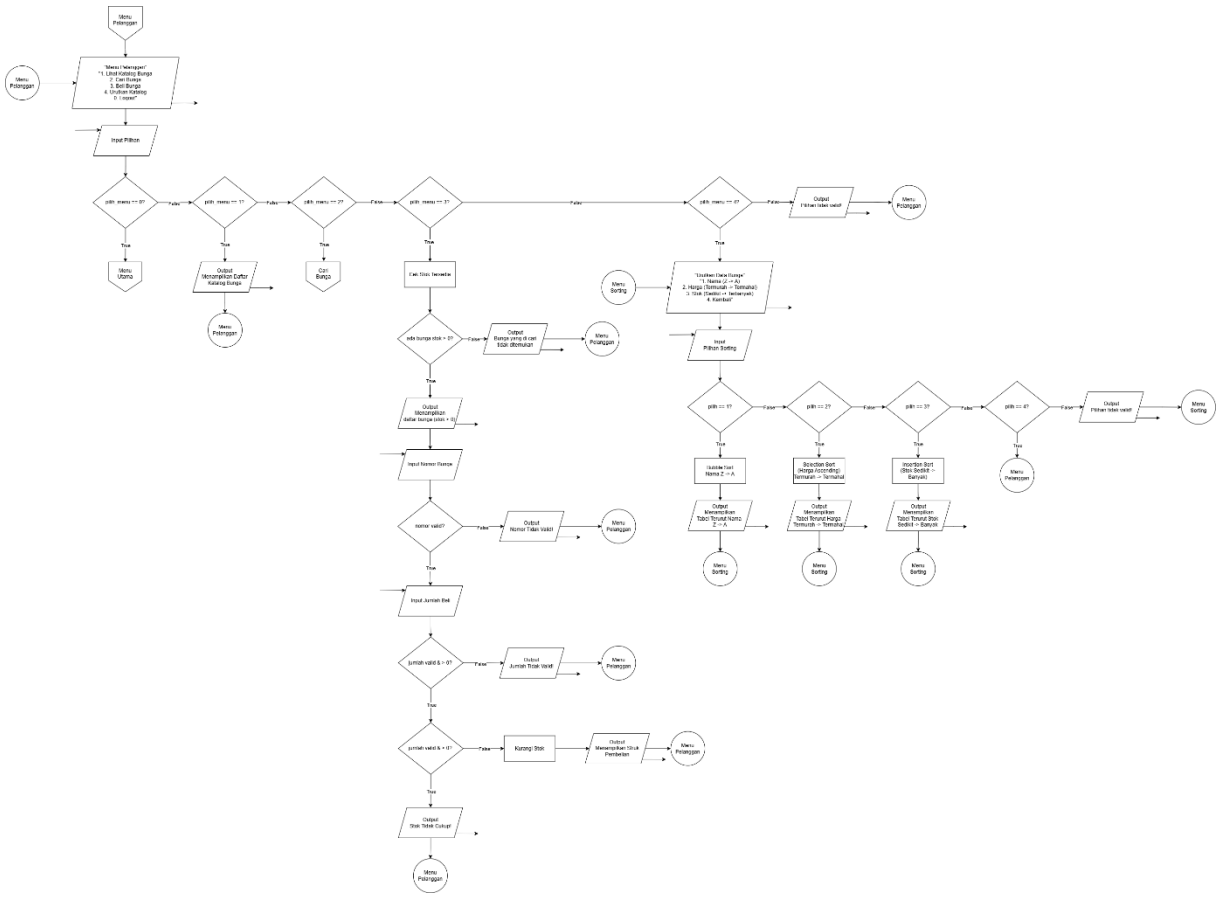
Flowchart mulai dengan bikin dua akun bawaan lalu masuk ke Menu Utama. Dari sini user bisa register untuk buat akun baru, login untuk masuk ke sistem, atau keluar untuk mengakhiri program. Kalau login berhasil, admin bisa kelola data bunga lewat menu CRUD, sedangkan pelanggan bisa lihat katalog,cari dan beli bunga. Kalau login gagal sebanyak 3 kali maka program langsung berhenti otomatis.



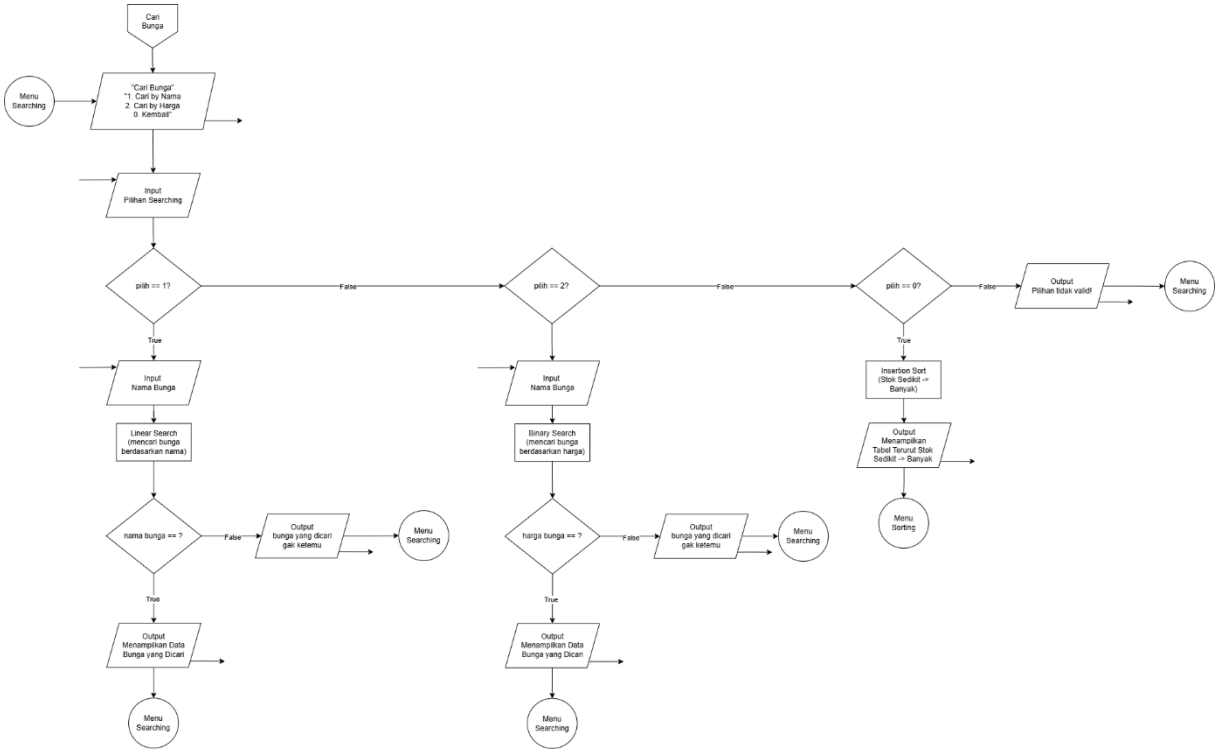
Gambar 1.1 Flowchart Menu Utama



Gambar 1.2 Flowchart Menu Admin



Gambar 1.3 Flowchart Menu Pelanggan



Gambar 1.4 Flowchart Menu Cari Bunga di Admin & Pelanggan

2. Deskripsi Singkat Program

Program ini bisa digunakan oleh dua jenis pengguna, admin yang ngurusin data bunganya, sama pelanggan yang mau beli atau cari bunga. Datanya menyimpan di memori selama program berjalan, jadi kalau programnya di tutup maka data tersebut hilang.

3. Souce Code

```
#include <iostream>
#include <iomanip>
#include <stdexcept>

using namespace std;

struct Harga {
    double satuan;
    string matauang;
};
struct Stok {
    int    jumlah;
    string satuan;
};
struct Bunga {
    string nama;
    Harga  harga;
    Stok   stok;
    string warna;
    int    aktif;
};
struct Pengguna {
    string nama;
    string password;
    string role;
    int    aktif;
};

int MAKS_PENGGUNA = 10;
int MAKS_BUNGA    = 100;

Pengguna dataPengguna[10];
Bunga    daftarBunga[100];

int jumlahPengguna = 0;
int totalBunga     = 0;

void menuAdmin(string namaUser);
void menuPelanggan(string namaUser);
int  doLogin(Pengguna* data, int jml);

string angkaKeStr(long long n) {
    if (n == 0) return "0";
    string s = "";
    while (n > 0) { s = char('0' + n % 10) + s; n /= 10; }
    return s;
}

string rupiahFormat(double angka) {
    string s = angkaKeStr((long long)angka);
    int n = s.size();
    string hasil = "";
    for (int i = 0; i < n; i++) {
```

```
        if (i > 0 && (n - i) % 3 == 0) hasil += ".";
        hasil += s[i];
    }
    return "Rp " + hasil;
}

int validAngka(string s) {
    if (s.size() == 0) return 0;
    for (int i = 0; i < (int)s.size(); i++)
        if (s[i] < '0' || s[i] > '9') return 0;
    return 1;
}

int bacaAngka(string s) {
    int hasil = 0;
    for (int i = 0; i < (int)s.size(); i++)
        hasil = hasil * 10 + (s[i] - '0');
    return hasil;
}

double bacaHarga(string s) {
    double h = 0;
    for (int i = 0; i < (int)s.size(); i++) h = h * 10 + (s[i] - '0');
    return h;
}

void bubbleSortNamaDesc(Bunga arr[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j].nama < arr[j + 1].nama) {
                Bunga tmp   = arr[j];
                arr[j]      = arr[j + 1];
                arr[j + 1]  = tmp;
            }
        }
    }
}

void selectionSortHargaAsc(Bunga arr[], int n) {
    for (int i = 0; i < n - 1; i++) {
        int idxMin = i;
        for (int j = i + 1; j < n; j++) {
            if (arr[j].harga.satuan < arr[idxMin].harga.satuan)
                idxMin = j;
        }
        if (idxMin != i) {
            Bunga tmp    = arr[i];
            arr[i]       = arr[idxMin];
            arr[idxMin]  = tmp;
        }
    }
}

void insertionSortStokAsc(Bunga arr[], int n) {
    for (int i = 1; i < n; i++) {
```

```
        Bunga kunci = arr[i];
        int j = i - 1;
        while (j >= 0 && arr[j].stok.jumlah > kunci.stok.jumlah) {
            arr[j + 1] = arr[j];
            j--;
        }
        arr[j + 1] = kunci;
    }
}

void tampilDenganSorting(int mode) {
    Bunga tmp[100];
    int jml = 0;
    for (int i = 0; i < totalBunga; i++)
        if (daftarBunga[i].aktif == 1) tmp[jml++] = daftarBunga[i];

    if (jml == 0) {
        cout << "Belum ada data bunga nih...\n";
        return;
    }

    if (mode == 1) bubbleSortNamaDesc(tmp, jml);
    else if (mode == 2) selectionSortHargaAsc(tmp, jml);
    else if (mode == 3) insertionSortStokAsc(tmp, jml);

    cout << "\n+----+--------------------+---------------+---------+-----------
--+\n";
    cout << "| No | Nama Bunga         | Harga         | Stok    | Warna
|\n";
    cout << "+----+--------------------+---------------+---------+------------
+\n";
    for (int i = 0; i < jml; i++) {
        cout << "| " << left << setw(3) << (i + 1)
             << "| " << setw(19) << tmp[i].nama
             << "| " << setw(15) << rupiahFormat(tmp[i].harga.satuan)
             << "| " << setw(8)  << tmp[i].stok.jumlah
             << "| " << setw(12) << tmp[i].warna << "|\n";
    }
    cout << "+----+--------------------+---------------+---------+------------
+\n";
}

void menuSorting() {
    string pilihan;
    while (true) {
        cout << "\n--- Urutkan Data Bunga ---\n";
        cout << "  [1] Nama (Z -> A)\n";
        cout << "  [2] Harga (Termurah -> Termahal)\n";
        cout << "  [3] Stok (Sedikit -> Terbanyak)\n";
        cout << "  [0] Kembali\n";
        cout << "Pilih: ";
        getline(cin, pilihan);

        if (pilihan == "1") {
            cout << "\n=== Urutkan Nama Z -> A ===";
```

```
            tampilDenganSorting(1);
        } else if (pilihan == "2") {
            cout << "\n=== Urutkan Harga Termurah -> Termahal ===";
            tampilDenganSorting(2);
        } else if (pilihan == "3") {
            cout << "\n=== Urutkan Stok Sedikit -> Terbanyak ===";
            tampilDenganSorting(3);
        } else if (pilihan == "0") {
            return;
        } else {
            cout << "Pilihan gak valid!\n";
        }
    }
}

int tampilTabel(int stockOnly) {
    cout << "\n+----+-------------------+---------------+---------+-----------
--+\n";
    cout << "| No | Nama Bunga        | Harga         | Stok    | Warna
|\n";
    cout << "+----+-------------------+---------------+---------+------------
+\n";

    int no = 0, adaData = 0;
    for (int i = 0; i < totalBunga; i++) {
        if (daftarBunga[i].aktif == 0) continue;
        if (stockOnly == 1 && daftarBunga[i].stok.jumlah == 0) continue;
        no++;
        adaData = 1;
        cout << "| " << left << setw(3) << no
             << "| " << setw(19) << daftarBunga[i].nama
             << "| " << setw(15) << rupiahFormat(daftarBunga[i].harga.satuan)
             << "| " << setw(8)  << daftarBunga[i].stok.jumlah
             << "| " << setw(12) << daftarBunga[i].warna << "|\n";
    }
    if (adaData == 0)
        cout << "|               (belum ada data bunga)
|\n";
    cout << "+----+-------------------+---------------+---------+------------
+\n";
    return no;
}

int pilihNomor(string aksi) {
    int peta[100], no = 0;
    for (int i = 0; i < totalBunga; i++)
        if (daftarBunga[i].aktif == 1) peta[no++] = i;

    tampilTabel(0);

    if (no == 0) { cout << "Yahh, Belum ada bunga nih...\n"; return -1; }

    string pilih;
    cout << "\nMau " << aksi << " bunga nomor berapa? ";
    getline(cin, pilih);
```

```
    if (validAngka(pilih) == 0) { cout << "Nomornya gak valid!\n"; return -1; }

    int nomor = bacaAngka(pilih);
    if (nomor < 1 || nomor > no) { cout << "Nomornya di luar jangkauan!\n";
return -1; }

    return peta[nomor - 1];
}

void tambahBunga() {
    cout << "\n--- Tambah Bunga Baru ---\n";

    try {
        if (totalBunga >= MAKS_BUNGA)
            throw runtime_error("Data bunga udah penuh nih!");

        Bunga bunga;
        string hargaStr, stokStr;

        cout << "Nama bunga  : "; getline(cin, bunga.nama);
        cout << "Harga (Rp)  : "; getline(cin, hargaStr);

        if (validAngka(hargaStr) == 0)
            throw invalid_argument("Harga sama stok harus angka ya!");

        cout << "Stok        : "; getline(cin, stokStr);

        if (validAngka(stokStr) == 0)
            throw invalid_argument("Harga sama stok harus angka ya!");

        cout << "Warna       : "; getline(cin, bunga.warna);

        bunga.harga.satuan   = bacaHarga(hargaStr);
        bunga.harga.matauang = "IDR";
        bunga.stok.jumlah    = bacaAngka(stokStr);
        bunga.stok.satuan    = "tangkai";
        bunga.aktif          = 1;

        daftarBunga[totalBunga++] = bunga;
        cout << "Bunga '" << bunga.nama << "' udah ditambahin!\n";

    } catch (const invalid_argument& e) {
        cout << e.what() << "\n";
    } catch (const runtime_error& e) {
        cout << e.what() << "\n";
    }
}

void lihatBunga() {
    cout << "\n--- Daftar Bunga di Toko Bunga Hias Rafalia ---";
    tampilTabel(0);
}

void prosesEditBunga(Bunga &bunga) {
```

```
    cout << "\nEdit '" << bunga.nama << "' (langsung Enter = gak berubah)\n\n";

    string namaBaru, hargaBaru, stokBaru, warnaBaru;

    cout << "Nama baru  [" << bunga.nama << "]: ";
    getline(cin, namaBaru);
    if (namaBaru.size() > 0) bunga.nama = namaBaru;

    cout << "Harga baru [" << rupiahFormat(bunga.harga.satuan) << "]: ";
    getline(cin, hargaBaru);
    if (hargaBaru.size() > 0) {
        if (validAngka(hargaBaru) == 1) bunga.harga.satuan =
bacaHarga(hargaBaru);
        else cout << "Harga gak valid, gak diubah.\n";
    }

    cout << "Stok baru  [" << bunga.stok.jumlah << "]: ";
    getline(cin, stokBaru);
    if (stokBaru.size() > 0) {
        if (validAngka(stokBaru) == 1) bunga.stok.jumlah = bacaAngka(stokBaru);
        else cout << "Stok gak valid, gak diubah.\n";
    }

    cout << "Warna baru [" << bunga.warna << "]: ";
    getline(cin, warnaBaru);
    if (warnaBaru.size() > 0) bunga.warna = warnaBaru;

    cout << "Data berhasil diupdate!\n";
}

void editBunga() {
    cout << "\n--- Edit Data Bunga ---\n";
    int idx = pilihNomor("edit");
    if (idx == -1) return;
    prosesEditBunga(daftarBunga[idx]);
}

void nonaktifkanBunga(int *statusAktif, string namaBunga) {
    *statusAktif = 0;
    cout << "Bunga '" << namaBunga << "' udah dihapus!\n";
}

void hapusBunga() {
    cout << "\n--- Hapus Bunga ---\n";
    int idx = pilihNomor("hapus");
    if (idx == -1) return;

    string konfirm;
    cout << "\nYakin mau hapus '" << daftarBunga[idx].nama << "'? (y/n): ";
    getline(cin, konfirm);

    if (konfirm == "y" || konfirm == "Y") {
        nonaktifkanBunga(&daftarBunga[idx].aktif, daftarBunga[idx].nama);
    } else {
        cout << "Penghapusan dibatalkan.\n";
```

```
    }
}

void lihatKatalog() {
    cout << "\n--- Katalog Bunga Rafalia ---";
    tampilTabel(0);
}

void cariNama() {
    cout << "\n--- Cari Bunga by Nama ---\n";

    try {
        int jml = 0;
        for (int i = 0; i < totalBunga; i++)
            if (daftarBunga[i].aktif == 1) jml++;
        if (jml == 0)
            throw runtime_error("Belum ada data bunga.");

        string keyword;
        cout << "Masukkan nama bunga yang dicari: ";
        getline(cin, keyword);

        if (keyword.empty())
            throw invalid_argument("Keywordnya gak boleh kosong dong!");

        for (int i = 0; i < (int)keyword.size(); i++)
            keyword[i] = tolower(keyword[i]);

        cout << "\n+----+--------------------+---------------+---------+-------
------+\n";
        cout << "| No | Nama Bunga         | Harga         | Stok    | Warna
|\n";
        cout << "+----+--------------------+---------------+---------+---------
----+\n";

        int ketemu = 0;
        for (int i = 0; i < totalBunga; i++) {
            if (daftarBunga[i].aktif == 0) continue;
            string sekarang = daftarBunga[i].nama;
            for (int j = 0; j < (int)sekarang.size(); j++)
                sekarang[j] = tolower(sekarang[j]);

            if (sekarang.find(keyword) != string::npos) {
                ketemu++;
                cout << "| " << left << setw(3) << ketemu
                     << "| " << setw(19) << daftarBunga[i].nama
                     << "| " << setw(15) << 
rupiahFormat(daftarBunga[i].harga.satuan)
                     << "| " << setw(8)  << daftarBunga[i].stok.jumlah
                     << "| " << setw(12) << daftarBunga[i].warna << "|\n";
            }
        }
        if (ketemu == 0)
            cout << "|         Bunga yang dicari gak ketemu...
|\n";
```

```
        cout << "+----+--------------------+----------------+---------+---------
----+\n";
        if (ketemu > 0) cout << "Ditemukan " << ketemu << " bunga.\n";

    } catch (const invalid_argument& e) {
        cout << e.what() << "\n";
    } catch (const runtime_error& e) {
        cout << e.what() << "\n";
    }
}

void cariHarga() {
    cout << "\n--- Cari Bunga by Harga ---\n";

    try {
        Bunga tmp[100];
        int jml = 0;
        for (int i = 0; i < totalBunga; i++)
            if (daftarBunga[i].aktif == 1) tmp[jml++] = daftarBunga[i];
        if (jml == 0)
            throw runtime_error("Belum ada data bunga.");

        for (int i = 0; i < jml - 1; i++) {
            int idxMin = i;
            for (int j = i + 1; j < jml; j++)
                if (tmp[j].harga.satuan < tmp[idxMin].harga.satuan)
                    idxMin = j;
            if (idxMin != i) {
                Bunga bantu  = tmp[i];
                tmp[i]       = tmp[idxMin];
                tmp[idxMin]  = bantu;
            }
        }

        string inputHarga;
        cout << "Masukkan harga yang dicari (Rp): ";
        getline(cin, inputHarga);

        if (validAngka(inputHarga) == 0)
            throw invalid_argument("Harganya harus angka ya!");

        double target = bacaHarga(inputHarga);
        int low = 0, high = jml - 1, posisi = -1;
        while (low <= high) {
            int mid = low + (high - low) / 2;
            if (tmp[mid].harga.satuan == target) {
                posisi = mid; break;
            } else if (tmp[mid].harga.satuan < target) {
                low = mid + 1;
            } else {
                high = mid - 1;
            }
        }

        if (posisi == -1)
```

```
            throw runtime_error("Bunga dengan harga " + rupiahFormat(target) + "
gak ada.");

        cout << "\n+----+--------------------+----------------+---------+-------
------+\n";
        cout << "| No | Nama Bunga         | Harga          | Stok    | Warna
|\n";
        cout << "+----+--------------------+----------------+---------+---------
----+\n";

        int no = 0;
        for (int i = 0; i < jml; i++) {
            if (tmp[i].harga.satuan == target) {
                no++;
                cout << "| " << left << setw(3) << no
                     << "| " << setw(19) << tmp[i].nama
                     << "| " << setw(15) << rupiahFormat(tmp[i].harga.satuan)
                     << "| " << setw(8)  << tmp[i].stok.jumlah
                     << "| " << setw(12) << tmp[i].warna << "|\n";
            }
        }
        cout << "+----+--------------------+----------------+---------+---------
----+\n";
        cout << "Ditemukan " << no << " bunga dengan harga " <<
rupiahFormat(target) << ".\n";

    } catch (const invalid_argument& e) {
        cout << e.what() << "\n";
    } catch (const runtime_error& e) {
        cout << e.what() << "\n";
    }
}

void menuCari() {
    string pilihan;
    while (true) {
        cout << "\n--- Cari Bunga ---\n";
        cout << "  [1] Cari by Nama\n";
        cout << "  [2] Cari by Harga\n";
        cout << "  [0] Kembali\n";
        cout << "Pilih: ";
        getline(cin, pilihan);

        if      (pilihan == "1") cariNama();
        else if (pilihan == "2") cariHarga();
        else if (pilihan == "0") return;
        else cout << "Pilihan gak valid!\n";
    }
}

void kurangiStok(int *stokPtr, int jumlah) {
    *stokPtr -= jumlah;
}

void beliBunga() {
```

```
    cout << "\n--- Beli Bunga ---\n";

    try {
        int adaStok = 0;
        for (int i = 0; i < totalBunga; i++)
            if (daftarBunga[i].aktif == 1 && daftarBunga[i].stok.jumlah > 0) {
adaStok = 1; break; }

        if (adaStok == 0)
            throw runtime_error("Maaf ya, semua bunga lagi habis nih...");

        int peta[100], no = 0;
        for (int i = 0; i < totalBunga; i++)
            if (daftarBunga[i].aktif == 1 && daftarBunga[i].stok.jumlah > 0)
                peta[no++] = i;

        tampilTabel(1);

        string pilihanBeli, pilihanJumlah;
        cout << "\nMau beli yang nomor berapa? ";
        getline(cin, pilihanBeli);

        if (validAngka(pilihanBeli) == 0 || bacaAngka(pilihanBeli) < 1 ||
bacaAngka(pilihanBeli) > no)
            throw invalid_argument("Nomornya gak valid!");

        int idx = peta[bacaAngka(pilihanBeli) - 1];
        Bunga& bunga = daftarBunga[idx];

        cout << "Mau beli berapa " << bunga.nama << "? (stok ada: " <<
bunga.stok.jumlah << "): ";
        getline(cin, pilihanJumlah);

        if (validAngka(pilihanJumlah) == 0 || bacaAngka(pilihanJumlah) <= 0)
            throw invalid_argument("Jumlahnya gak valid!");

        int jumlah = bacaAngka(pilihanJumlah);
        if (jumlah > bunga.stok.jumlah)
            throw runtime_error("Stok gak cukup! Cuma ada " +
angkaKeStr(bunga.stok.jumlah) + " " + bunga.stok.satuan + ".");

        double total = bunga.harga.satuan * jumlah;
        kurangiStok(&bunga.stok.jumlah, jumlah);

        cout << "\n--- Pembelian Berhasil! ---\n";
        cout << "  Bunga    : " << bunga.nama << "\n";
        cout << "  Warna    : " << bunga.warna << "\n";
        cout << "  Harga    : " << rupiahFormat(bunga.harga.satuan) << " / " <<
bunga.stok.satuan << "\n";
        cout << "  Jumlah   : " << jumlah << " " << bunga.stok.satuan << "\n";
        cout << "  --------------------------------\n";
        cout << "  Total    : " << rupiahFormat(total) << "\n";
        cout << "  Makasih udah belanja di Toko Bunga Hias Rafalia!\n";

    } catch (const invalid_argument& e) {
```

```
        cout << e.what() << "\n";
    } catch (const runtime_error& e) {
        cout << e.what() << "\n";
    }
}

void doRegister() {
    if (jumlahPengguna >= MAKS_PENGGUNA) {
        cout << "Yah, slot pengguna udah penuh nih!\n";
        return;
    }

    cout << "\n=== DAFTAR AKUN BARU ===\n";

    try {
        string nama, password;

        cout << "Nama : ";
        getline(cin, nama);

        for (int i = 0; i < jumlahPengguna; i++) {
            if (dataPengguna[i].nama == nama)
                throw runtime_error("Yah, nama itu udah dipake orang lain, pake
nama lain ya!...");
        }

        cout << "Password : ";
        getline(cin, password);

        if (password.empty())
            throw invalid_argument("Password gak boleh kosong!");

        dataPengguna[jumlahPengguna].nama     = nama;
        dataPengguna[jumlahPengguna].password = password;
        dataPengguna[jumlahPengguna].role     = "pelanggan";
        dataPengguna[jumlahPengguna].aktif    = 1;
        jumlahPengguna++;

        cout << "\nYes! Akun '" << nama << "' berhasil dibuat sebagai
pelanggan!\n";

    } catch (const invalid_argument& e) {
        cout << e.what() << "\n";
    } catch (const runtime_error& e) {
        cout << e.what() << "\n";
    }
}

int doLogin(Pengguna* data, int jml) {
    int coba = 0;
    while (coba < 3) {
        cout << "\n=== LOGIN TOKO BUNGA HIAS RAFALIA ===\n";

        try {
            string inputNama, inputPassword;
```

```
            cout << "Nama : "; getline(cin, inputNama);
            cout << "Password  : "; getline(cin, inputPassword);

            int ketemu = -1;
            for (int i = 0; i < jml; i++) {
                if (data[i].aktif == 1 &&
                    data[i].nama     == inputNama &&
                    data[i].password == inputPassword) {
                    ketemu = i;
                    break;
                }
            }

            if (ketemu == -1)
                throw runtime_error("Nama atau Password salah.");

            cout << "\nLogin berhasil! Hai " << data[ketemu].nama << " :)\n";
            if (data[ketemu].role == "admin")
                menuAdmin(data[ketemu].nama);
            else
                menuPelanggan(data[ketemu].nama);
            return 1;

        } catch (const runtime_error& e) {
            coba++;
            cout << "\nLogin gagal. " << e.what() << "\n";
            if (coba < 3)
                cout << "Kesempatan mencoba sisa " << (3 - coba) << " lagi\n";
        }
    }

    cout << "\nTerlalu banyak percobaan login. Program berhenti.\n";
    return 0;
}

void menuAdmin(string namaUser) {
    string pilihan;
    while (true) {
        cout << "\n==================================================\n";
        cout << "       TOKO BUNGA HIAS RAFALIA - ADMIN\n";
        cout << "==================================================\n";
        cout << "  Halo, " << namaUser << "!\n\n";
        cout << "  [1] Tambah Bunga Baru\n";
        cout << "  [2] Lihat Semua Bunga\n";
        cout << "  [3] Edit Data Bunga\n";
        cout << "  [4] Hapus Bunga\n";
        cout << "  [5] Urutkan Data Bunga\n";
        cout << "  [6] Cari Bunga\n";
        cout << "  [0] Logout\n";
        cout << "==================================================\n";
        cout << "  Mau ngapain? (0-6): ";
        getline(cin, pilihan);

        if      (pilihan == "1") tambahBunga();
        else if (pilihan == "2") lihatBunga();
```

```
        else if (pilihan == "3") editBunga();
        else if (pilihan == "4") hapusBunga();
        else if (pilihan == "5") menuSorting();
        else if (pilihan == "6") menuCari();
        else if (pilihan == "0") { cout << "Sampai jumpa lagi! BYE BYE\n";
return; }
        else cout << "Pilihan gak valid, coba lagi dong...\n";
    }
}

void menuPelanggan(string namaUser) {
    string pilihan;
    while (true) {
        cout << "\n==================================================\n";
        cout << "     TOKO BUNGA HIAS RAFALIA - PELANGGAN\n";
        cout << "==================================================\n";
        cout << "  Halo, " << namaUser << "!\n\n";
        cout << "  [1] Lihat Katalog Bunga\n";
        cout << "  [2] Cari Bunga\n";
        cout << "  [3] Beli Bunga\n";
        cout << "  [4] Urutkan Katalog\n";
        cout << "  [0] Keluar\n";
        cout << "==================================================\n";
        cout << "  Mau apa? (0-4): ";
        getline(cin, pilihan);

        if      (pilihan == "1") lihatKatalog();
        else if (pilihan == "2") menuCari();
        else if (pilihan == "3") beliBunga();
        else if (pilihan == "4") menuSorting();
        else if (pilihan == "0") { cout << "Makasih udah mampir! Sampai jumpa
lagi ya :)\n"; return; }
        else cout << "Pilihan gak valid...\n";
    }
}

int main() {

    dataPengguna[jumlahPengguna].nama     = "Rafa";
    dataPengguna[jumlahPengguna].password = "104";
    dataPengguna[jumlahPengguna].role     = "admin";
    dataPengguna[jumlahPengguna].aktif    = 1;
    jumlahPengguna++;

    dataPengguna[jumlahPengguna].nama     = "Tamu";
    dataPengguna[jumlahPengguna].password = "000";
    dataPengguna[jumlahPengguna].role     = "pelanggan";
    dataPengguna[jumlahPengguna].aktif    = 1;
    jumlahPengguna++;

    cout << "\n=======================================================\n";
    cout << "|     Selamat Datang di Toko Bunga Hias Rafalia      |\n";
    cout << "=======================================================\n";

    string pilihan;
```

```
    int berjalan = 1;
    while (berjalan) {
        cout << "\n=====================================================\n";
        cout << "|             MENU UTAMA TOKO BUNGA HIAS             |\n";
        cout << "=====================================================\n";
        cout << "|  [1] Register  (Daftar Akun Baru)                  |\n";
        cout << "|  [2] Login     (Masuk ke Akun)                     |\n";
        cout << "|  [0] Keluar                                        |\n";
        cout << "=====================================================\n";
        cout << "Pilih menu (0-2): ";
        getline(cin, pilihan);

        if      (pilihan == "1") doRegister();
        else if (pilihan == "2") {
            int hasil = doLogin(dataPengguna, jumlahPengguna);
            if (hasil == 0) berjalan = 0;
        }
        else if (pilihan == "0") {
            cout << "\nTerima kasih udah mampir ke Toko Bunga Hias
Rafalia!\n\n";
            berjalan = 0;
        }
        else cout << "Pilihan tidak valid. Silakan pilih 0-2.\n";
    }

    return 0;
}
```

Gambar 1.5 Source Code Program

4. Hasil Output

```
=========================================================
|     Selamat Datang di Toko Bunga Hias Rafalia      |
=========================================================

=========================================================
|             MENU UTAMA TOKO BUNGA HIAS             |
=========================================================
|  [1] Register  (Daftar Akun Baru)                  |
|  [2] Login     (Masuk ke Akun)                     |
|  [0] Keluar                                        |
=========================================================
Pilih menu (0-2):
```

Gambar 1.6 Output Menu Utama

```
=====================================================
     TOKO BUNGA HIAS RAFALIA - PELANGGAN
=====================================================
  Halo, ancel!

  [1] Lihat Katalog Bunga
  [2] Cari Bunga
  [3] Beli Bunga
  [4] Urutkan Katalog
  [0] Keluar
=====================================================
  Mau apa? (0-4):
```

Gambar 1.7 Output Menu Pelanggan

```
==================================================
     TOKO BUNGA HIAS RAFALIA - PELANGGAN
==================================================
  Halo, ancel!

  [1] Lihat Katalog Bunga
  [2] Cari Bunga Favorit
  [3] Beli Bunga
  [4] Urutkan Katalog
  [0] Keluar
==================================================
  Mau apa? (0-4): 1

--- Katalog Bunga Rafalia ---
+----+--------------------+----------------+---------+-------------+
| No | Nama Bunga         | Harga          | Stok    | Warna       |
+----+--------------------+----------------+---------+-------------+
| 1  | matahari           | Rp 40.000      | 40      | kuning      |
| 2  | melati             | Rp 10.000      | 10      | putih       |
| 3  | bunga mawar        | Rp 30.000      | 30      | merah       |
| 4  | bunga tulip        | Rp 50.000      | 50      | pink        |
+----+--------------------+----------------+---------+-------------+
```

Gambar 1.8 Output Lihat Katalog Bunga

```
  --- Cari Bunga ---
    [1] Cari by Nama
    [2] Cari by Harga
    [0] Kembali
  Pilih: 1

  --- Cari Bunga by Nama ---
  Masukkan nama bunga yang dicari: tulip

  +----+--------------------+----------------+---------+-------------+
  | No | Nama Bunga         | Harga          | Stok    | Warna       |
  +----+--------------------+----------------+---------+-------------+
  | 1  | tulip              | Rp 50.000      | 50      | pink        |
  +----+--------------------+----------------+---------+-------------+
  Ditemukan 1 bunga.

  --- Cari Bunga ---
    [1] Cari by Nama
    [2] Cari by Harga
    [0] Kembali
  Pilih:
```

Gambar 1.9 Output Cari Bunga Linear Search

```
--- Cari Bunga ---
  [1] Cari by Nama
  [2] Cari by Harga
  [0] Kembali
Pilih: 2

--- Cari Bunga by Harga ---
Masukkan harga yang dicari (Rp): 30000

+----+--------------------+----------------+---------+-------------+
| No | Nama Bunga         | Harga          | Stok    | Warna       |
+----+--------------------+----------------+---------+-------------+
| 1  | matahari           | Rp 30.000      | 30      | kuning      |
+----+--------------------+----------------+---------+-------------+
Ditemukan 1 bunga dengan harga Rp 30.000.

--- Cari Bunga ---
  [1] Cari by Nama
  [2] Cari by Harga
  [0] Kembali
Pilih:
```

Gambar 1.10 Output Cari Bunga Binary Search

```
--- Beli Bunga ---

+----+--------------------+----------------+---------+-------------+
| No | Nama Bunga         | Harga          | Stok    | Warna       |
+----+--------------------+----------------+---------+-------------+
| 1  | matahari           | Rp 40.000      | 40      | kuning      |
| 2  | melati             | Rp 10.000      | 10      | putih       |
| 3  | bunga mawar        | Rp 30.000      | 30      | merah       |
| 4  | bunga tulip        | Rp 50.000      | 50      | pink        |
+----+--------------------+----------------+---------+-------------+

Mau beli yang nomor berapa? 4
Mau beli berapa bunga tulip? (stok ada: 50): 40

--- Pembelian Berhasil! ---
  Bunga    : bunga tulip
  Warna    : pink
  Harga    : Rp 50.000 / tangkai
  Jumlah   : 40 tangkai
  --------------------------------
  Total    : Rp 2.000.000
  Makasih udah belanja di Toko Bunga Hias Rafalia!
```

Gambar 1.11 Output Beli Bunga

```
--- Urutkan Data Bunga ---
  [1] Nama (Z -> A)
  [2] Harga (Termurah -> Termahal)
  [3] Stok (Sedikit -> Terbanyak)
  [0] Kembali
Pilih:
```

Gambar 1.12 Output Urutkan Data Bunga

```
=== Urutkan Nama Z -> A ===
+----+--------------------+----------------+---------+-------------+
| No | Nama Bunga         | Harga          | Stok    | Warna       |
+----+--------------------+----------------+---------+-------------+
| 1  | melati             | Rp 10.000      | 10      | putih       |
| 2  | matahari           | Rp 40.000      | 40      | kuning      |
| 3  | bunga tulip        | Rp 50.000      | 10      | pink        |
| 4  | bunga mawar        | Rp 30.000      | 30      | merah       |
+----+--------------------+----------------+---------+-------------+
```

Gambar 1.13 Output Urutkan Nama Z-A

```
=== Urutkan Harga Termurah -> Termahal ===
+----+--------------------+----------------+---------+-------------+
| No | Nama Bunga         | Harga          | Stok    | Warna       |
+----+--------------------+----------------+---------+-------------+
| 1  | melati             | Rp 10.000      | 10      | putih       |
| 2  | bunga mawar        | Rp 30.000      | 30      | merah       |
| 3  | matahari           | Rp 40.000      | 40      | kuning      |
| 4  | bunga tulip        | Rp 50.000      | 10      | pink        |
+----+--------------------+----------------+---------+-------------+
```

Gambar 1.14 Output Urutkan Harga Termurah-Termahal

```
=== Urutkan Stok Sedikit -> Terbanyak ===
+----+--------------------+----------------+---------+-------------+
| No | Nama Bunga         | Harga          | Stok    | Warna       |
+----+--------------------+----------------+---------+-------------+
| 1  | melati             | Rp 10.000      | 10      | putih       |
| 2  | bunga tulip        | Rp 50.000      | 10      | pink        |
| 3  | bunga mawar        | Rp 30.000      | 30      | merah       |
| 4  | matahari           | Rp 40.000      | 40      | kuning      |
+----+--------------------+----------------+---------+-------------+
```

Gambar 1.15 Output Urutkan Stok Sedikit-Terbanyak

```
--- Beli Bunga ---

+----+--------------------+----------------+---------+-------------+
| No | Nama Bunga         | Harga          | Stok    | Warna       |
+----+--------------------+----------------+---------+-------------+
| 1  | matahari           | Rp 40.000      | 40      | kuning      |
| 2  | melati             | Rp 10.000      | 10      | putih       |
| 3  | bunga mawar        | Rp 30.000      | 30      | merah       |
| 4  | bunga tulip        | Rp 50.000      | 50      | pink        |
+----+--------------------+----------------+---------+-------------+

Mau beli yang nomor berapa? 4
Mau beli berapa bunga tulip? (stok ada: 50): 40

--- Pembelian Berhasil! ---
  Bunga    : bunga tulip
  Warna    : pink
  Harga    : Rp 50.000 / tangkai
  Jumlah   : 40 tangkai
  -------------------------------
  Total    : Rp 2.000.000
  Makasih udah belanja di Toko Bunga Hias Rafalia!
```

Gambar 1.16 Output Tampilan Menu Admin

```
==================================================
       TOKO BUNGA HIAS RAFALIA - ADMIN
==================================================
  Halo, Rafa!

  [1] Tambah Bunga Baru
  [2] Lihat Semua Bunga
  [3] Edit Data Bunga
  [4] Hapus Bunga
  [5] Urutkan Data Bunga
  [6] Cari Bunga
  [0] Logout
==================================================
  Mau ngapain? (0-6): 1

--- Tambah Bunga Baru ---
Nama bunga  : lily
Harga (Rp)  : 40000
Stok        : 40
Warna       : ungu
Bunga 'lily' udah ditambahin!
```

Gambar 1.17 Output Tambah Bunga Baru

```
==================================================
       TOKO BUNGA HIAS RAFALIA - ADMIN
==================================================
  Halo, Rafa!

  [1] Tambah Bunga Baru
  [2] Lihat Semua Bunga
  [3] Edit Data Bunga
  [4] Hapus Bunga
  [5] Urutkan Data Bunga
  [6] Cari Bunga
  [0] Logout
==================================================
  Mau ngapain? (0-6): 2

--- Daftar Bunga di Toko Bunga Hias Rafalia ---
+----+--------------------+----------------+---------+-------------+
| No | Nama Bunga         | Harga          | Stok    | Warna       |
+----+--------------------+----------------+---------+-------------+
| 1  | matahari           | Rp 30.000      | 30      | kuning      |
| 2  | tulip              | Rp 50.000      | 50      | pink        |
| 3  | mawar              | Rp 20.000      | 20      | merah       |
| 4  | melati             | Rp 10.000      | 10      | putih       |
| 5  | lily               | Rp 40.000      | 40      | ungu        |
+----+--------------------+----------------+---------+-------------+
```

Gambar 1.18 Output Lihat Semua Bunga

```
--- Edit Data Bunga ---

+----+--------------------+----------------+---------+-------------+
| No | Nama Bunga         | Harga          | Stok    | Warna       |
+----+--------------------+----------------+---------+-------------+
| 1  | matahari           | Rp 30.000      | 30      | kuning      |
| 2  | tulip              | Rp 50.000      | 50      | pink        |
| 3  | mawar              | Rp 20.000      | 20      | merah       |
| 4  | melati             | Rp 10.000      | 10      | putih       |
| 5  | lily               | Rp 40.000      | 40      | ungu        |
+----+--------------------+----------------+---------+-------------+

Mau edit bunga nomor berapa? 5

Edit 'lily' (langsung Enter = gak berubah)

Nama baru  [lily]:
Harga baru [Rp 40.000]: 45000
Stok baru  [40]:
Warna baru [ungu]:
Data berhasil diupdate!
```

Gambar 1.19 Output Edit Data Bunga

```
--- Hapus Bunga ---

+----+--------------------+----------------+---------+-------------+
| No | Nama Bunga         | Harga          | Stok    | Warna       |
+----+--------------------+----------------+---------+-------------+
| 1  | matahari           | Rp 30.000      | 30      | kuning      |
| 2  | tulip              | Rp 50.000      | 50      | pink        |
| 3  | mawar              | Rp 20.000      | 20      | merah       |
| 4  | melati             | Rp 10.000      | 10      | putih       |
| 5  | lily               | Rp 45.000      | 40      | ungu        |
+----+--------------------+----------------+---------+-------------+

Mau hapus bunga nomor berapa? 5

Yakin mau hapus 'lily'? (y/n): y
Bunga 'lily' udah dihapus!
```

Gambar 1.20 Output Hapus Bunga

```
==================================================
        TOKO BUNGA HIAS RAFALIA - ADMIN
==================================================
  Halo, Rafa!

  [1] Tambah Bunga Baru
  [2] Lihat Semua Bunga
  [3] Edit Data Bunga
  [4] Hapus Bunga
  [5] Urutkan Data Bunga
  [6] Cari Bunga
  [0] Logout
==================================================
  Mau ngapain? (0-6): 5

--- Urutkan Data Bunga ---
  [1] Nama (Z -> A)
  [2] Harga (Termurah -> Termahal)
  [3] Stok (Sedikit -> Terbanyak)
  [0] Kembali
Pilih:
```

Gambar 1.21 Output Pilihan 5 Menu Sorting

```
=== Urutkan Nama Z -> A ===
+----+--------------------+----------------+---------+-------------+
| No | Nama Bunga         | Harga          | Stok    | Warna       |
+----+--------------------+----------------+---------+-------------+
| 1  | melati             | Rp 10.000      | 10      | putih       |
| 2  | matahari           | Rp 40.000      | 40      | kuning      |
| 3  | bunga tulip        | Rp 50.000      | 50      | pink        |
| 4  | bunga mawar        | Rp 30.000      | 30      | merah       |
+----+--------------------+----------------+---------+-------------+
```

Gambar 1.22 Output Pilihan 1 Urutkan Nama Z – A

```
=== Urutkan Harga Termurah -> Termahal ===
+----+--------------------+----------------+---------+-------------+
| No | Nama Bunga         | Harga          | Stok    | Warna       |
+----+--------------------+----------------+---------+-------------+
| 1  | melati             | Rp 10.000      | 10      | putih       |
| 2  | bunga mawar        | Rp 30.000      | 30      | merah       |
| 3  | matahari           | Rp 40.000      | 40      | kuning      |
| 4  | bunga tulip        | Rp 50.000      | 50      | pink        |
+----+--------------------+----------------+---------+-------------+
```

Gambar 1.23 Output Pilihan 2 Urutkan Harga Termurah – Termahal

```
=== Urutkan Stok Sedikit -> Terbanyak ===
+----+--------------------+----------------+---------+-------------+
| No | Nama Bunga         | Harga          | Stok    | Warna       |
+----+--------------------+----------------+---------+-------------+
| 1  | melati             | Rp 10.000      | 10      | putih       |
| 2  | bunga mawar        | Rp 30.000      | 30      | merah       |
| 3  | matahari           | Rp 40.000      | 40      | kuning      |
| 4  | bunga tulip        | Rp 50.000      | 50      | pink        |
+----+--------------------+----------------+---------+-------------+
```

Gambar 1.24 Output Pilihan 3 Urutkan Stok Sedikit – Terbanyak

```
--- Cari Bunga ---
  [1] Cari by Nama
  [2] Cari by Harga
  [0] Kembali
Pilih: 1

--- Cari Bunga by Nama ---
Masukkan nama bunga yang dicari: tulip

+----+--------------------+----------------+---------+-------------+
| No | Nama Bunga         | Harga          | Stok    | Warna       |
+----+--------------------+----------------+---------+-------------+
| 1  | tulip              | Rp 50.000      | 50      | pink        |
+----+--------------------+----------------+---------+-------------+
Ditemukan 1 bunga.

--- Cari Bunga ---
  [1] Cari by Nama
  [2] Cari by Harga
  [0] Kembali
Pilih:
```

Gambar 1.25 Output Menu Cari Bunga di Admin & Pelanggan

```
--- Cari Bunga ---
  [1] Cari by Nama
  [2] Cari by Harga
  [0] Kembali
Pilih: 1

--- Cari Bunga by Nama ---
Masukkan nama bunga yang dicari: tulip

+----+--------------------+----------------+---------+-------------+
| No | Nama Bunga         | Harga          | Stok    | Warna       |
+----+--------------------+----------------+---------+-------------+
| 1  | tulip              | Rp 50.000      | 50      | pink        |
+----+--------------------+----------------+---------+-------------+
Ditemukan 1 bunga.

--- Cari Bunga ---
  [1] Cari by Nama
  [2] Cari by Harga
  [0] Kembali
Pilih:
```

Gambar 1.26 Output Linear Scearch

```
--- Cari Bunga by Harga ---
Masukkan harga yang dicari (Rp): 40000

+----+--------------------+----------------+---------+-------------+
| No | Nama Bunga         | Harga          | Stok    | Warna       |
+----+--------------------+----------------+---------+-------------+
| 1  | lily               | Rp 40.000      | 40      | ungu        |
+----+--------------------+----------------+---------+-------------+
Ditemukan 1 bunga dengan harga Rp 40.000.

--- Cari Bunga ---
  [1] Cari by Nama
  [2] Cari by Harga
  [0] Kembali
Pilih:
```

Gambar 1.27 Output Linear Scearch

5. Langkah - Langkah GIT

5.1 GIT INIT

```
PS C:\Users\lenovoindo\OneDrive\Documents\Praktikum-APD> git init
Reinitialized existing Git repository in C:/Users/lenovoindo/OneDrive/Documents/Praktikum-APD/.git/
```

Untuk membuat repositori Git baru di sebuah direktori proyek, baik itu proyek yang baru atau proyek yang sudah ada dan belum dikelola oleh Git.

5.2 GIT ADD

```
PS C:\Users\lenovoindo\OneDrive\Documents\Praktikum-APD> git add .
```

perintah di Git untuk menyiapkan atau "staging" perubahan file Anda untuk dimasukkan ke dalam riwayat proyek (commit berikutnya).

5.3 GIT COMMIT

```
PS C:\Users\lenovoindo\OneDrive\Documents\Praktikum-APD\post-test\post-test-3> git commit -m "post test 3"
On branch main
nothing to commit, working tree clean
```

Untuk menyimpan (merekam) perubahan yang telah Anda buat pada file ke dalam riwayat proyek Anda, menciptakan "snapshot" dari proyek pada titik waktu tertentu.

5.4 GIT REMOTE

```
PS C:\Users\lenovoindo\OneDrive\Documents\Praktikum-APD\post-test\post-test-3> git remote add origin https://github.com/anza-cyber/Praktikum
-APD-c1.git
```

Untuk berbagi perubahan kode, mengelola versi proyek, dan berkolaborasi secara efektif melalui operasi push dan pull.

5.5 GIT PUSH

```
PS C:\Users\lenovoindo\OneDrive\Documents\Praktikum-APD\post-test\post-test-3> git push -u origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 2.00 KiB | 293.00 KiB/s, done.
Total 7 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/anza-cyber/Praktikum-APD-c1.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\lenovoindo\OneDrive\Documents\Praktikum-APD\post-test\post-test-3>
```

Untuk mengunggah perubahan dari repositori lokal Anda ke repositori jarak jauh (seperti GitHub), sehingga perubahan tersebut dapat dibagikan dengan orang lain atau disinkronkan dengan versi proyek yang lebih besar.